

Capstone Dataset and Baseline

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Model Cards and Human Review

Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Subgroup Performance

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Define fields for Capstone data and requirements. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone data and requirements. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and baseline procedure. When may final labels be consulted?

6. Solve the given case

Capstone has 60 independent leaves, several photos per leaf, healthy/spotted labels.

Plan grouping and baseline.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Your capstone dataset includes duplicate photographs in both training and validation. What should be repaired before measuring the baseline?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 34 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and baseline procedure. When may final labels be consulted?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 State a question

Write a bounded question and a non-use statement for the supplied task.

My reasoning and evidence

13. W2 Define fields

Explain why record ID, group ID and measured width have different roles.

My reasoning and evidence

14. W3 Set criteria

Name three acceptance criteria that do not require the candidate to win.

My reasoning and evidence

15. W4 Apply independently

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

My reasoning and evidence

16. W5 Prepare a fallback

What should the team do if it cannot obtain a valid untouched final test?

My reasoning and evidence

17. Capstone models and baselines — W1 Build the baseline

State the worked training-majority rule and validation accuracy.

My reasoning and evidence

18. Capstone models and baselines — W2 Evaluate the candidate

Compute the worked candidate accuracy, precision and recall.

My reasoning and evidence

19. Capstone models and baselines — W3 Explain the tradeoff

Compare errors rather than only headline accuracy.

My reasoning and evidence

20. Capstone models and baselines — W4 Apply

independently

Training 18 clear/6 blocked; validation baseline[[9, 0],[3, 0]], candidate[[8, 1],[1, 2]]. Calculate accuracy for each and candidate precision/recall.

My reasoning and evidence

21. Capstone models and baselines — W5 Freeze evidence

List what must be saved before the teacher releases final-test labels.

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Capstone Dataset and Baseline

Plan the plant capstone around independent groups and a baseline before collecting more rows.

Toy proposed dataset: 8 plants, 3 photos each. Development groups A–F; final groups G, H. Target is visible-spot status, with review for hidden leaves.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Capstone Dataset and Baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

Write visible-spot and review rules before annotation.



2. Split by groups

Keep all 18 A-F views in development and all 6 G/H views for final evaluation.



3. Plan baseline

Fit majority label only from development training; record metrics and error types.

CHANGE ONE THING

A derived crop of plantG is added to training.

Explain the changed result using the matching lesson.

What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

24. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan the plant capstone around independent groups and a baseline before collecting more rows.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Fit majority label only from development training; record metrics and error types.

1. Define target
2. Split by groups
3. Plan baseline

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Plan baseline Fit majority label only from development training; record metrics and error types. Read the picture in order. Fit majority label only from development training; record metrics and error types. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: A derived crop of plantG is added to training. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The final group separation is violated.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The final group separation is violated. The crop retains the same source plant identity even with a new file name.

Repair a missing building block

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match evidence. Short practice: Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read training-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
